Online Retail UK-Business Intelligence & Customer Analyst

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Sales, Customer Segmentation, Retention & Lifetime Analysis

Executive Summary

- Key Finding:
  - Total Revenue reached **£991.7M**, with significant Q4 seasonal spike.
  - Revenue is highly concentrated: a small segment of high-value customers contributes a disproportionate share of total sales.
  - Average monthly retention rate is **37%,** with notable drop-off after first few months.
- Business Implications:
  - Revenue stability depends heavily on retaining high-value customers.
  - Early-stage churn significantly impacts long-term customer lifetime value.
  - Inventory and marketing planning must align with Q4 demand surge.
- Limitation Note:
  - Dataset covers Dec 2010 – Dec 2011 (2010 partial month).

Business Context

Project Goal
To transform raw transaction-level data into **actionable business insights** that support **revenue growth** and **customer retention** strategy.

Sales Performance	How is overall sales performing over time?	Total Sales, Monthly Trend, YoY Comparison
Seasonality Patterns	Are there seasonal patterns in sales?	Monthly Sales %, Q4 Contribution
Product Performance	Which products contribute most to revenue?	Top 10 Products, Revenue Contribution
Customer Segmentation	Who are the most valuable customers?	High / Medium / Low Value Segments
Customer Retention	Are customers returning over time?	Monthly Retention Rate, Cohort Analysis
Customer Lifetime	How long and how valuable are customers?	Lifetime Value, Repeat Behavior

Dataset Overview

Dataset : Online Retail UK
Period : 2010-2011
Transactions-based data

Inside the data:
- 500,000 transaction records
- Invoice-level granularity
- December 2010 – December 2011
- Data cleaning applied (canceled invoice removed)

Key fields:
- InvoiceDate
- Customer ID
- Quantity
- UnitPrice
- Country

Data Preparation

- Removed canceled transactions
- Standardized date format
- Created revenue column (Quantity × Unit Price)
- Handled missing Customer IDs

KPI Definition

- Total Revenue = SUM(Quantity × Unit Price)
- Retention Rate = Returning Customers / Previous Month Customers
- Customer Lifetime = Months between first and last purchase

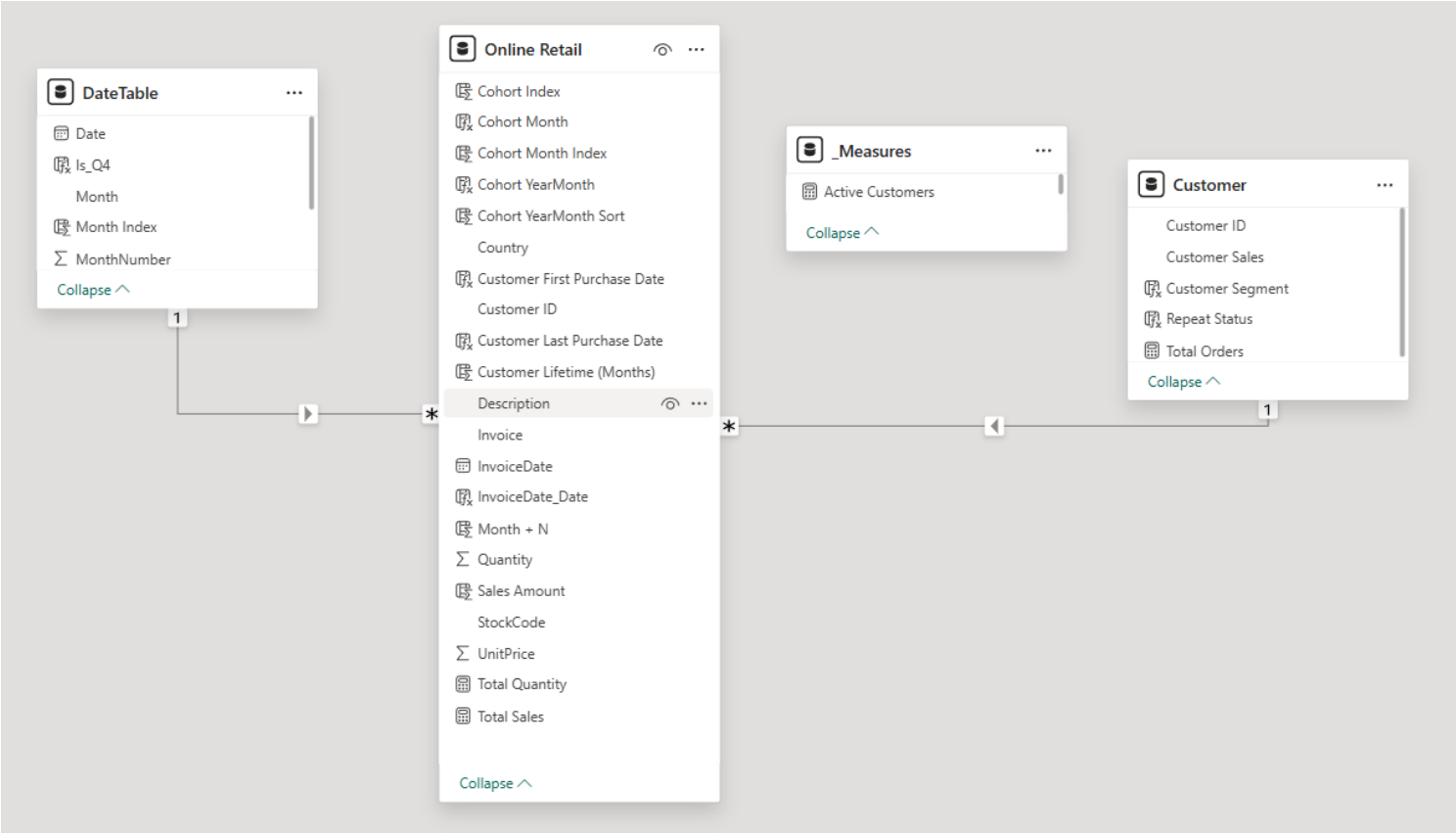
Data Modeling Strategy

- Star-schema inspired structure
- Created separate Date Table
- Defined relationships for time intelligence

Analytical Techniques Applied

- Time Intelligence (MoM, Seasonality)
- Revenue Contribution Analysis
- Cohort Analysis
- Customer Lifetime Analysis

Data Modeling



Star-schema inspired model

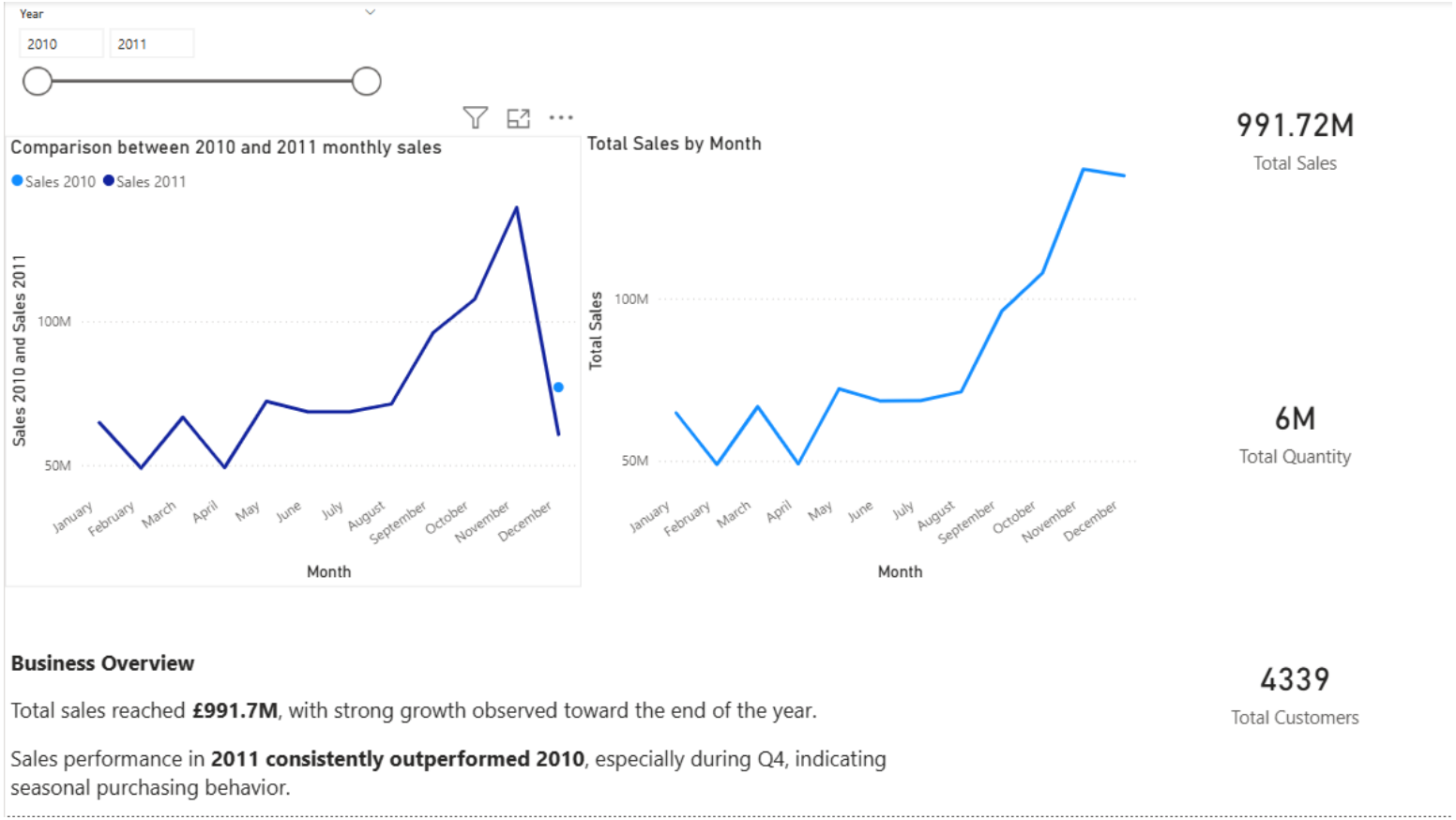
Fact Table : Online Retail
Dimension Table :

- DateTable
- Customer

Established relationships to ensure accurate aggregation.

Optimized model for DAX performance and clarity.

Dashboard Overview



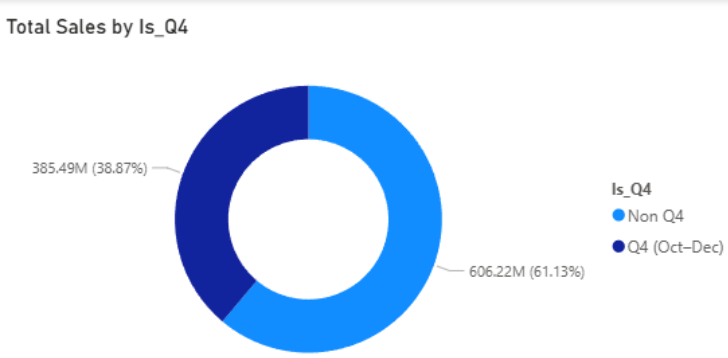
Key KPI:

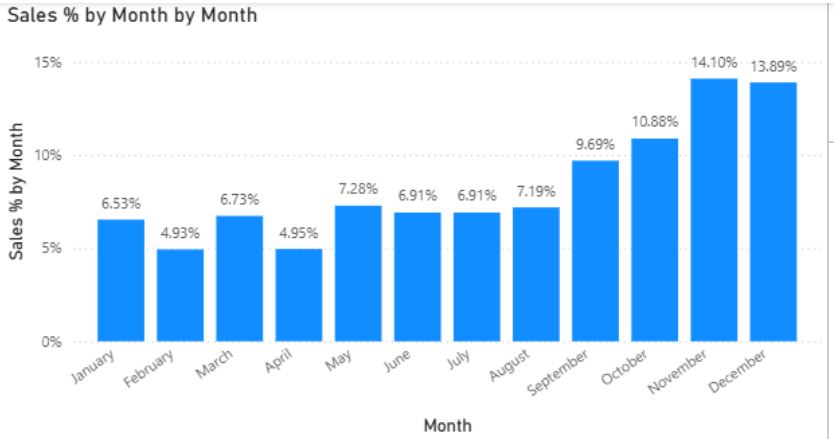
- Total Sales
- Total Quantity
- Sales Trend over Time
- Year-over-Year Comparison

Insight Summary

- Sales show upward trend with strong end-of-year performance
- Clear growth momentum towards Q4

Seasonality Analysis





Seasonality Insight

Nearly **39% of annual sales occur in Q4**, confirming strong holiday-driven demand.

Sales steadily increase toward year-end, peaking in **November–December**, which highlights the importance of inventory and marketing readiness during this period.

Highest Monthly Sales Share	Q4 Contribution to Annual Sales
0.14	0.39
Peak Month Sales %	Q4 Sales %

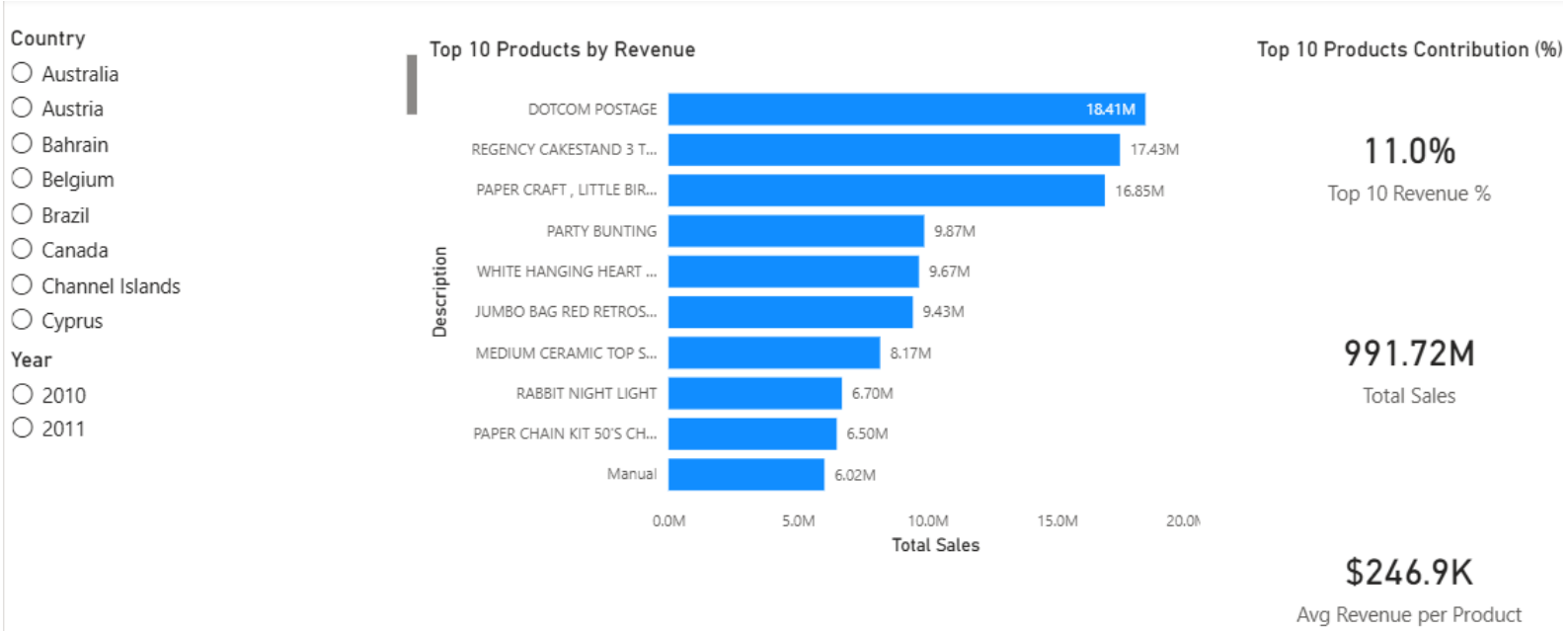
Key Findings:

- Q4 contributes a significant portion of annual sales
- December consistently shows peak sales
- Sales distribution is not uniform across months

Business Impact

- Inventory and marketing planning should prioritize Q4
- Promotional strategies can be aligned with peak periods

Product Performance Analysis



Product Performance Insight

Revenue is highly concentrated among a small group of products.

The top 10 products contribute a significant share of total sales, indicating opportunities for focused inventory optimization and cross-selling strategies.

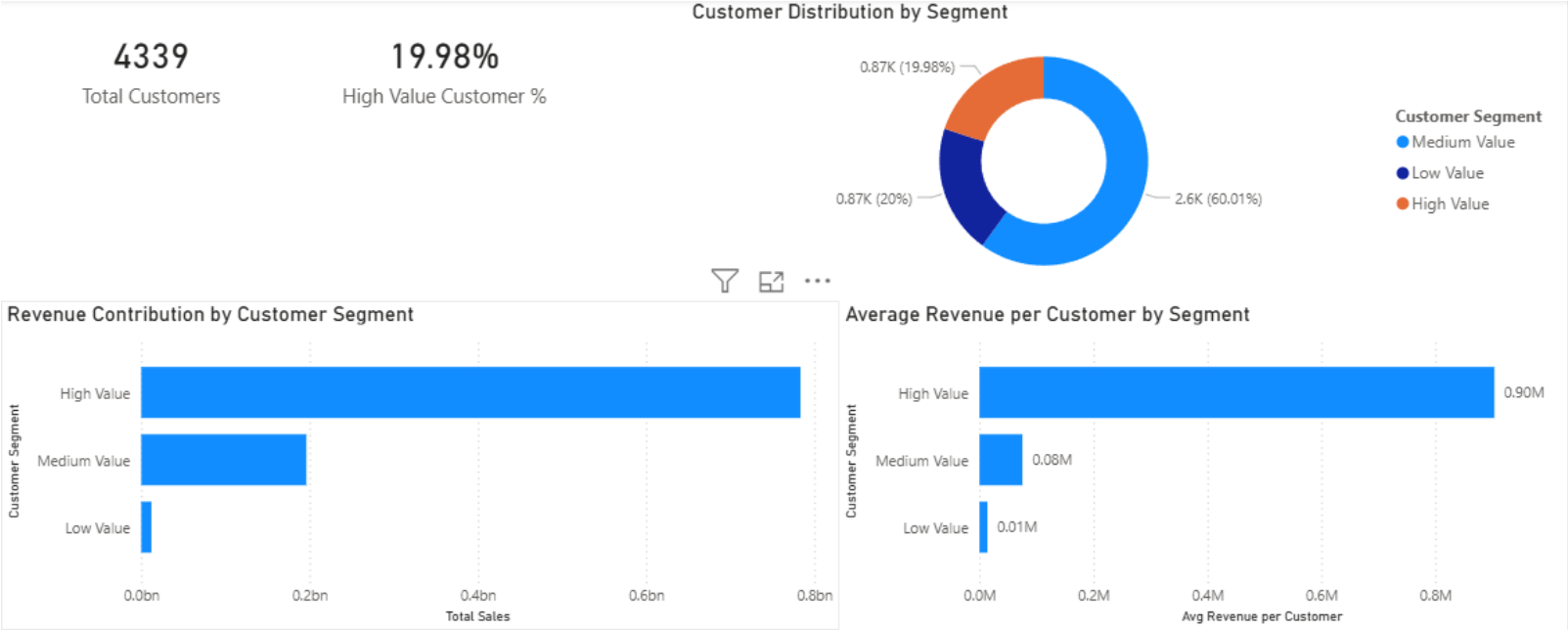
Key Insights:

- Revenue is highly concentrated in top products
- Top 10 products generate a large share of total sales
- Long tail of low-performing products exists

Opportunity:

- Optimize product portfolio
- Focus marketing and inventory on high-performing items

Customer Segmentation



Although High Value customers represent only around 20% of the total customer base, they contribute the majority of total revenue. This indicates a strong revenue concentration among a small group of loyal and high-spending customers.
Medium Value customers form the largest segment by volume, suggesting potential opportunities to increase revenue through targeted upselling and retention strategies.
In addition, High Value customers show significantly higher average revenue per customer, reinforcing their importance as the primary revenue drivers for the business.

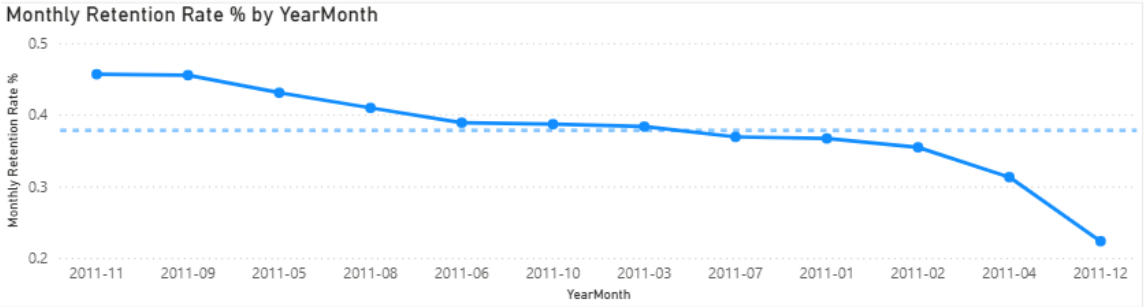
Customers segmented into:

- High Value
- Medium Value
- Low Value

Key Findings:

- Small percentage of customers generate majority of revenue
- High Value customers are critical for business sustainability

Customer Retention & Cohort Analysis



"Based on last available month (Dec 2011)"

0.65
Avg Monthly Churn Rate %

616
Active Customers (Last Month)

22.34%
Retention Rate (Last Month)

Sum of Year	Month	Active Customers	Active Customers PM	Retained Customers	Monthly Retention Rate %
62341	January	742	886	325	0.37
56308	February	759	742	263	0.35
62341	March	975	759	291	0.38
60330	April	857	975	305	0.31
62341	May	1057	857	369	0.43
60330	June	992	1057	411	0.39
62341	July	950	992	366	0.37
62341	August	936	950	389	0.41
60330	September	1267	936	426	0.46
62341	October	1365	1267	490	0.39
60330	November	1665	1365	623	0.46
80409	December	1266	1665	647	0.39

Year	0	1	2	3	4	5	6	7	8	9	10	11	12	Total
2010	100.00%	36.68%	32.39%	38.49%	36.34%	39.84%	36.34%	34.99%	35.44%	39.62%	37.47%	50.34%	26.64%	**100.00%**
2011	100.00%	18.88%	19.00%	17.00%	16.91%	15.23%	13.99%	12.05%	10.63%	8.43%	5.15%	1.42%		**100.00%**
Total	**100.00%**	**22.52%**	**21.73%**	**21.39%**	**20.88%**	**20.26%**	**18.55%**	**16.73%**	**15.69%**	**14.80%**	**11.75%**	**11.41%**	**5.44%**	**100.00%**

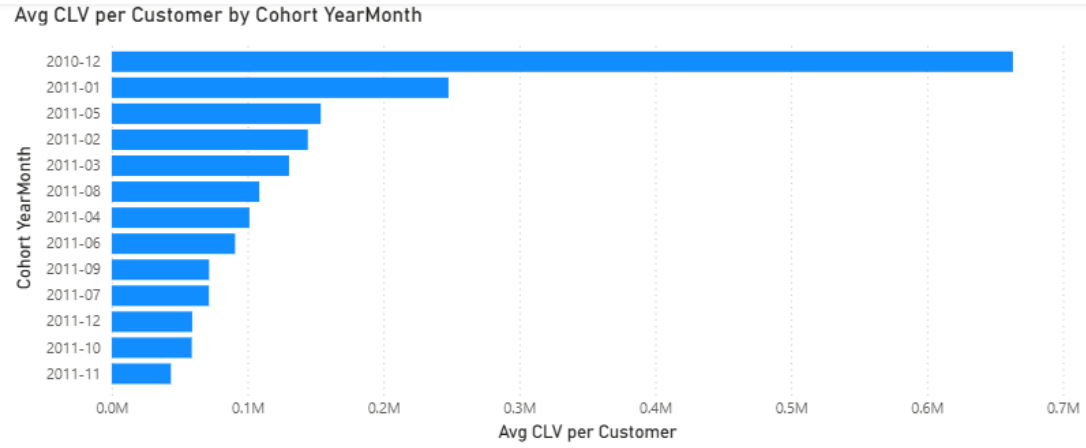
Retention Analysis:

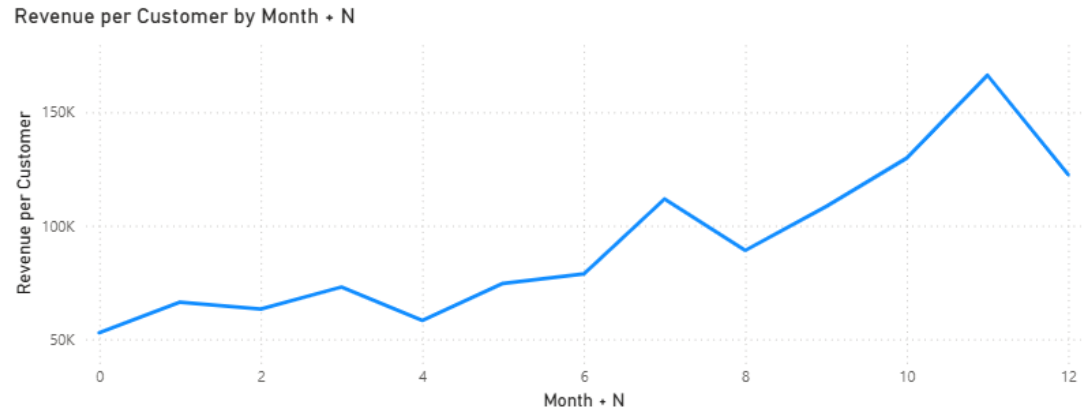
- Monthly active and retained customers tracked
- Cohort analysis used to measure repeat behavior over time

Insights:

- Retention declines gradually over months
- First few months are critical for customer loyalty

Customer Lifetime Insights





228.56K
Avg Revenue per Customer

4.60
Avg Orders per Customer

13.00
Avg Customer Lifetime (Months)

Lifetime Analysis:

- Customer lifetime measured in months
- Repeat customers show longer lifetimes and higher value

Key Insights:

- Retention directly impacts long-term revenue
- Investing in customer loyalty yields compounding returns

BUSINESS RECOMMENDATIONS

01.

Prioritize

Prioritize retention campaigns in early customer lifecycle

02.

Focus

Focus on High Value customer segment & personalized promotions

03.

Align

Increase stock allocation and marketing budget ahead of peak season

04.

Optimize

Prioritize supply chain and promotion strategy around top performing products.

Technical Skills

Power BI, DAX (Time Intelligence, Cohort, Retention), Data Modeling & Data Visualization

Analytical Skills

Business problem framing, Insight generation, Data storytelling

ASSUMPTIONS & LIMITATIONS

This project demonstrates an end-to-end Business Intelligence workflow, from raw transaction data to actionable business insights:

- Dataset limited to UK retail transactions
- No marketing or cost data available
- Customer behavior inferred from purchase history only
- Analysis is intended for analytical demonstration purposes.

Thank You